



Executive
Perspectives

Outperforming Through Volatility: AI in Refining

Refining

April 2026

Introduction

We meet often with CEOs to discuss AI, a topic that is both fascinating and rapidly evolving. After working with more than 2,000 clients over the past two+ years, we are sharing our latest insights in a new series designed to help CEOs navigate AI. For refining leaders, the question is no longer whether AI matters, but how to use it to outperform through volatility and create durable competitive advantage.

The refining sector is under structural pressure. Long-term demand is expected to decline, while net refining capacity could still grow by 2030, even after closures, sustaining pressure on margins. At the same time, geopolitics, sanctions, regulation, and trade dislocations are amplifying volatility. AI offers a new set of levers—moving refineries from periodic, siloed optimization to real-time, margin-driven decisioning and closed-loop workflows across planning, operations, logistics, maintenance, and safety. **For a reference mid-quartile player, that can translate into ~\$0.5–1.2/bbl of EBIT uplift by 2030.** Against this backdrop, three questions matter most:

- What does it mean to operate as an AI-first refinery end to end?
- Where are leading refiners already creating value today in planning and scheduling, operations, logistics and trading, maintenance and reliability, turnarounds, and HSSE?
- What pragmatic steps should companies take to move from isolated pilots to scaled transformation?

Advances in AI, generative AI, and agentic AI are enabling refiners to shift from **isolated tools to embedded capabilities**

In this Executive Perspective, we outline how AI-first refineries can outperform in a structurally challenged and volatile market



Executive summary | AI-first refineries will outperform in a structurally challenged and volatile market

WHY

AI adoption is now critical for refineries facing structural pressures

- The **refining outlook is tightening**, long-term demand is expected to decline, while net refining capacity **could grow by 2030**—even after closures—sustaining pressure on margins. **Geopolitics, sanctions, regulation**, and trade dislocations are **amplifying volatility**.
- At the same time, **recent advances in AI and computing are opening new value pools** that were previously inaccessible. For a reference mid-quartile player, AI can **unlock ~\$0.5–1.2/bbl of EBIT uplift** by 2030.

WHAT

AI is already shaping core refinery workflows, from optimization to autonomous decisioning

- **AI is moving refineries** from periodic, siloed optimization to **real-time, margin-driven decisioning and closed-loop workflows** across planning, operations, logistics, maintenance, and safety.
- The **impact is already visible** across the value chain: **planning and scheduling can deliver ~\$0.15–0.30/bbl**, **operations ~\$0.1–0.5/bbl**, logistics/trading ~\$0.2–0.3/bbl, and maintenance/reliability ~\$0.05–0.1/bb.
- **Proven deployments show material results, including +\$80M from a ~300 kbpd refinery** in operations, >\$80M annual margin opportunity in logistics/trading, and **5%–15% maintenance cost reduction** with 62% inspection-efficiency gains.

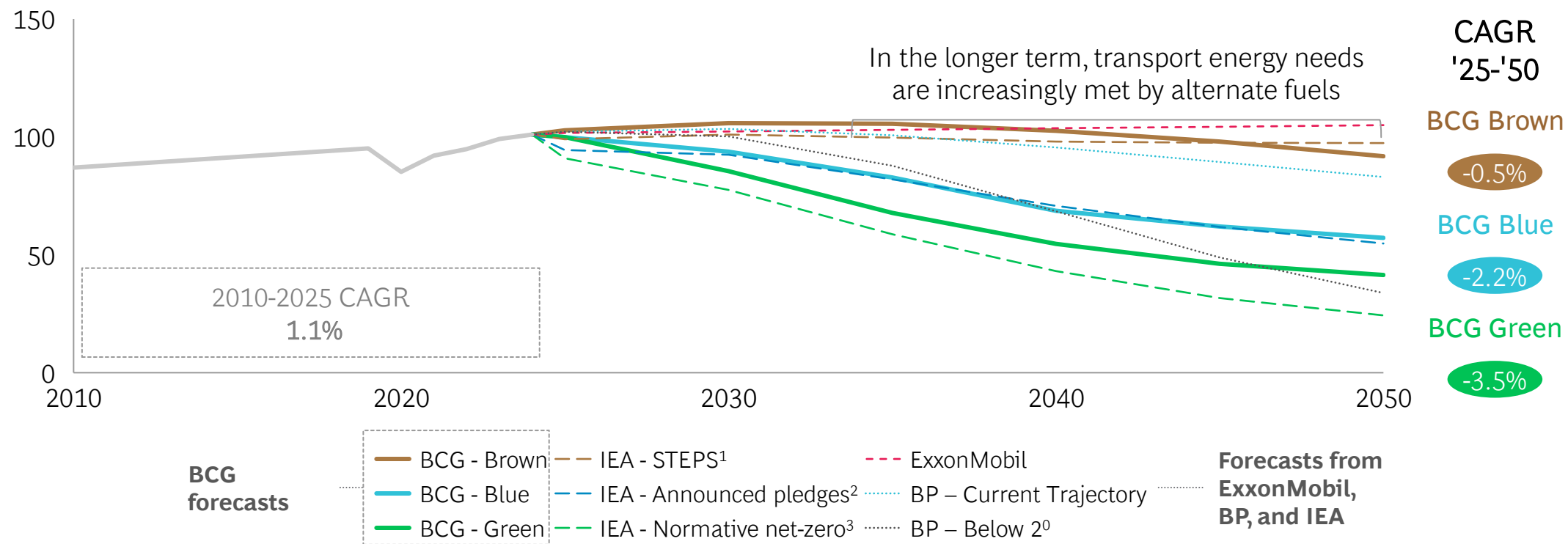
HOW

Winning refineries anchor AI in P&L impact through focused formations

- **Start with the highest-value workflows**—planning and scheduling, operations, and maintenance and reliability—where AI directly improves margin and reduces value leakage.
- **Build the foundations to scale**: standardized data, digital twins, enterprise platforms, and workflow integration are critical to embed AI into day-to-day execution.
- **Redesign processes and the operating model**—don't just deploy tools—to capture value at scale, with governance, value tracking, and capability building embedded from day one.

Long-term oil demand is set to decline across a spectrum of scenarios

Refined products global demand (Mbpd)

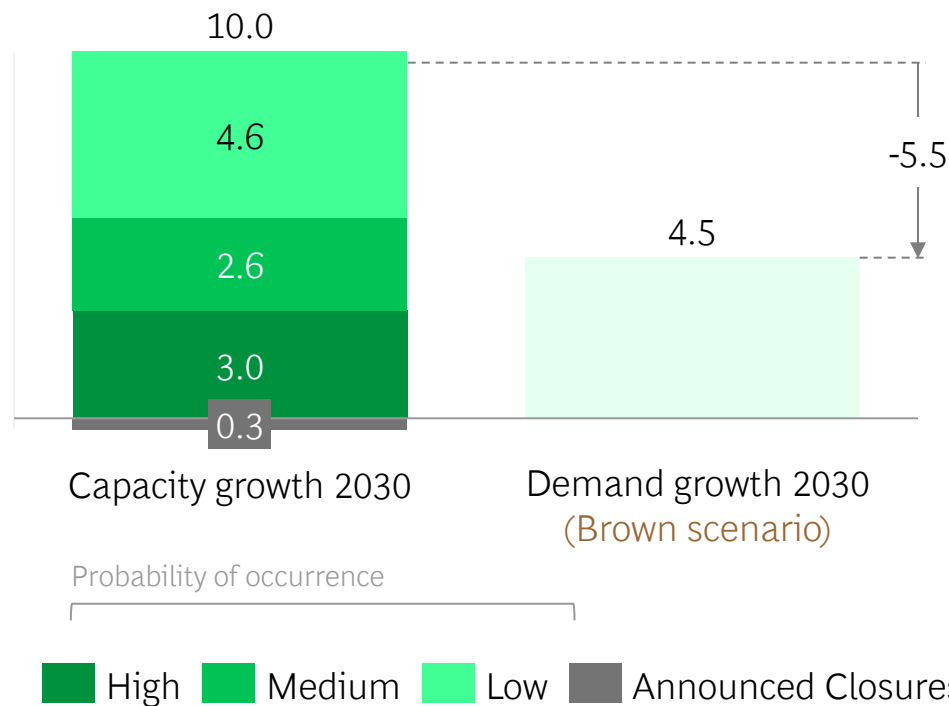


Actual demand and mix will be defined by future regulations and how pledges translate into policies

1. IEA STEP scenario. 2. IEA APS scenario, emission reduction is from 2022 to 2050. 3. Scenario slope of IEA NZE has been adjusted in order to reflect a stronger emission reduction effort from 2030. Temperature rises and CO2 reduction for cases 2 and 3 are approximate and derived by linear interpolation from IEA STEPS and IEA APS scenarios, not bottom-up climate modeling. Source: IEA WEO 2024; IEA WEO 2025; BP; ExxonMobil; BCG Oil Long Term Model; BCG analysis.

Refinery capacity growth is set to outpace demand, putting sustained pressure on the margins

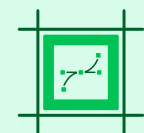
Net refining capacity growth 2026-2030^{1,2,3} (Mbpd)



Supply exceeds demand even in the conservative Brown scenario, where demand decline is minimal



Even if we exclude capacities with "low" probability, supply exceeds most optimistic demand growth scenario



Supply-demand dynamics de-averaged by refined product (e.g., jet fuel, naphtha demand more resilient)

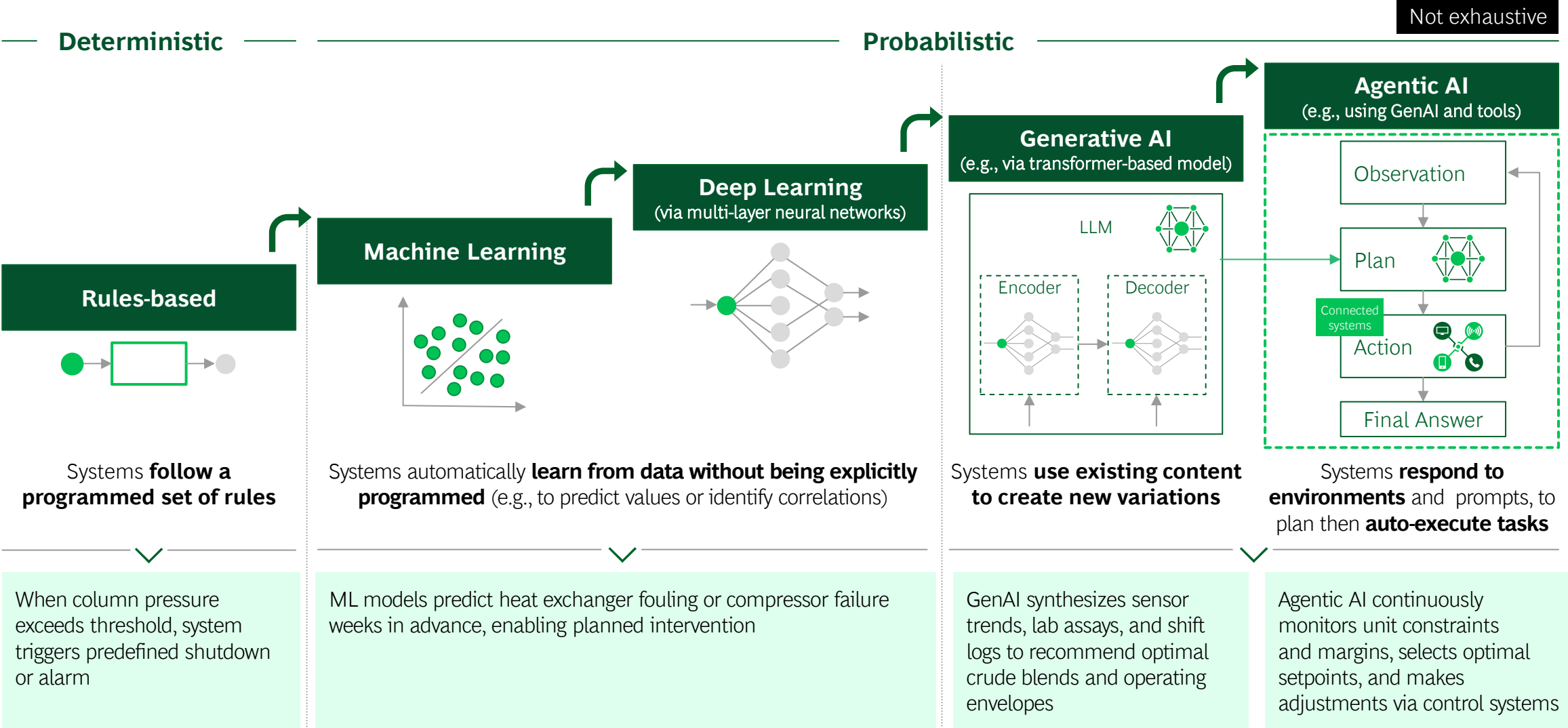
1. Not probability weighted. 2. CDU + Condensate. 3. Including both expansion projects of existing refineries and new refineries, including projects that roll over from 2025, and excluding capacity added or closed in 2025.
Note: Excluding greenfield refinery projects under approval or prior stages. Probability category attribution subject to available information at the time of elaboration of the slide.
Source: GlobalData 2025; IEA WEO 2025; Wood Mackenzie; BCG Oil Long Term Model; BCG Global Refining Model; BCG Analysis.

However, margins are expected to be highly volatile in the near term

						
	Electrification of transport	Chinese exports	Global growth and tariffs	Evolving Russian sanctions	New capacity startups	IMO MEPC83 ¹
Market dynamic	Faster EV and hybrid uptake...	Potentially higher export quotas ...	Uneven economic growth and trade barriers across regions ...	US sanctions on Rosneft/Lukoil and the EU's 2026 product ban ...	Accelerating startups of large-scale refining capacity....	Tightening emissions standards ² across the global fleet...
Margin impact	... erodes gasoline demand and caps margin upside	... pressure regional margins through oversupply and price competition	...create margin volatility from uneven demand and product flows	...potentially disrupt trade and amplify margin volatility	... exacerbates oversupply and compresses margins	... require operational reinvestments with penalties for non-compliance

1. International Maritime Organization Marine Environment Protection Committee. 2. Carbon intensity cuts of 20%–30% by 2030 and 70%–80% by 2040 vs 2008 levels. Source: BCG analysis

Rapid advances in AI and high-performance computing have the potential to fundamentally transform the refining industry



AI is already driving a step-change in the efficiency and efficacy of core functions of the refining value chain

Not exhaustive



Planning and Scheduling



Operations



Logistics and Trading



Maintenance and Reliability



Turnarounds and Projects



HSSE¹

1. Health, Safety, Sustainability, and Environment.
Note: Business support functions are excluded.
Source: BCG analysis

From...

- **Periodic constraint-based planning** using LP models and offline simulations, with limited response to real-time disruptions
- Advanced process control **optimizing individual units with manual, cross-unit coordination**
- **Conservative blending** and **manual quality checks**
- Deterministic planning with static **demand and inventory assumptions** with reliance on **buffers**
- Predictive **maintenance based on historical data and expert judgement**
- Capex decisions based on **deterministic business cases and static schedules, updated periodically**
- Safety managed through **audits, procedures, and lagging/ leading indicators**
- **Rules-based energy management** and emissions tracking **driven by expert judgment**

To...

- **Real-time, AI-driven planning continuously reoptimizing** production, logistics, and product slate using live plant and market data
- **Autonomous, plant-wide closed-loop optimization**, AI-driven blending **optimizing** quality, yield, and cost **in real time**
- AI-enabled stochastic optimization **dynamically balancing inventories and logistics** across scenarios to maximize margin
- AI agents **orchestrating maintenance end-to-end** predicting failures and prioritizing interventions
- AI **continuously stress-testing scope and sequencing across thousands of scenarios** identifying optimal paths much more quickly
- AI-enabled **continuous risk sensing and incident prediction**
- **Model-driven, scenario-based optimization** under real-time constraints

AI is unlocking several high-impact applications across the refining value chain

1 Planning and Scheduling

AI-driven crude slate and margin optimizer

 AI-based short-interval scheduling optimizer

 Automated plan-to-actual root cause diagnosis and learning

Automated multi-scenario refinery planning and feasibility

2 Operations

 AI-assisted real-time optimization, control room

Lab and quality analytics for spec giveaway prevention

 AI energy management optimizer

3 Logistics and Trading

AI inventory reconciliation and loss detection

Integrated logistics network and transport optimization

 Market structure prediction for proactive trades

4 Maintenance and Reliability

Predictive maintenance platform

 Risk-based inspection and integrity analytics

AI maintenance planning and scheduling optimizer

AI-driven equipment strategy optimizer

5 Turnarounds and Projects

 Integrated turnaround planning and resource scheduler

Turnaround execution control tower

AI-assisted inspection planning and execution

MOC and compliance co-pilot

6 HSSE

 Predictive safety risk and incident detection

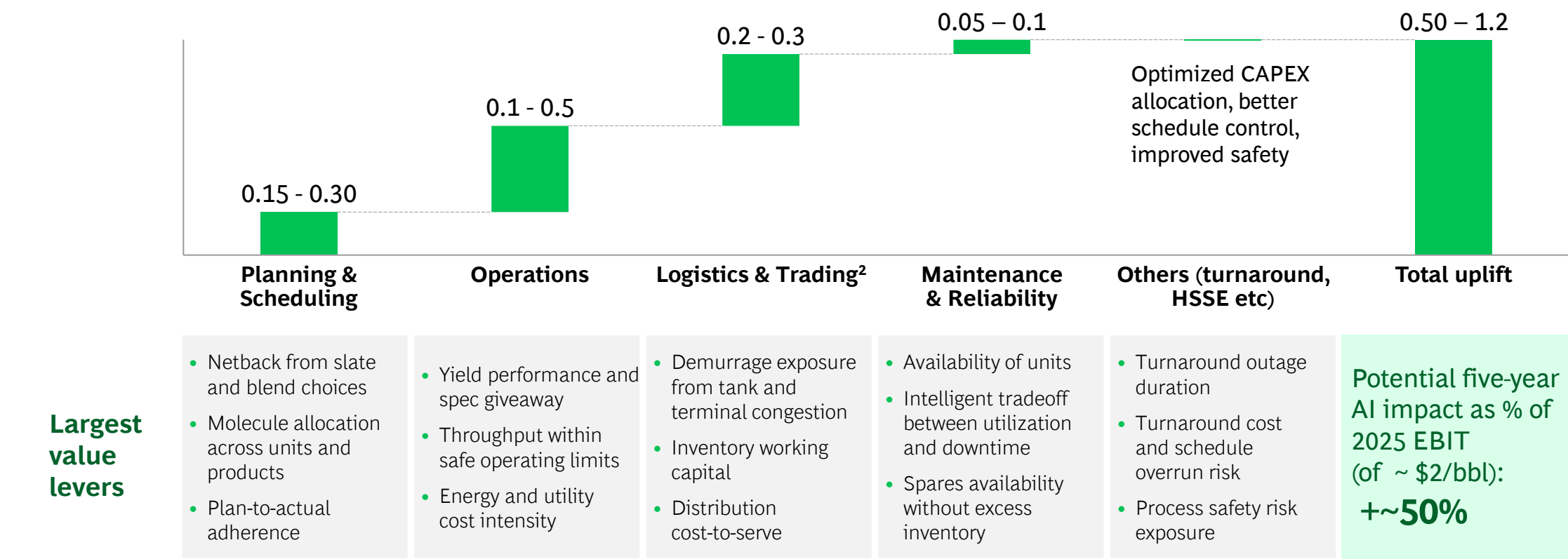
Carbon intensity analytics, reporting automation

Common business support functions (legal, HR, IT, procurement)

Material opportunities exist in these functions as well, but are not Chemicals-specific and therefore out of scope of this report

AI has potential to improve refining EBIT by over 50% by 2030

2030 AI impact on EBIT¹ (\$/bbl)



1. For a reference player- mid-quartile. Examples of AI-levers in refining: gross margin (e.g. AI-driven real-time prescriptive input of optimal operating points, maximizing throughputs and yields value and optimal scheduling plan generation with AI algorithms), energy and utilities (e.g., real-time utility network monitoring and optimization to minimize energy and H2 cost), maintenance (e.g., LLM-based agents to support maintenance operators in end-to-end execution including work order generation, image-based unit diagnostics, repair instructions, and log and report automation), CapEx run and maintenance (e.g., capital deployment optimization based on risk-based optimization of project selection supported by AI algorithms). 2. Logistics, maintenance, G&A.

AI improves margin and operational performance through adaptive scheduling that rapidly reoptimizes under changing conditions

The challenge

At a Southeast Asian refiner, short-term scheduling was constrained by:

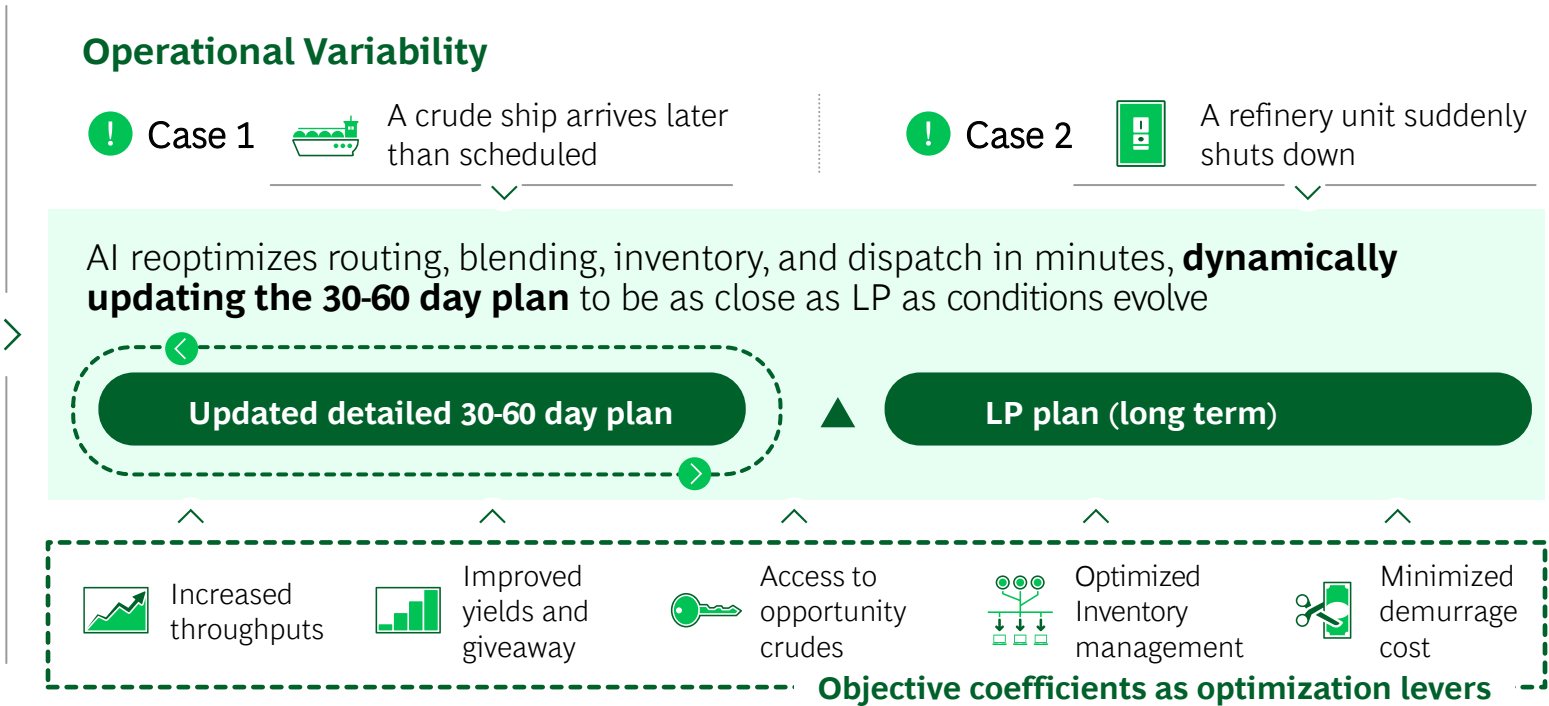
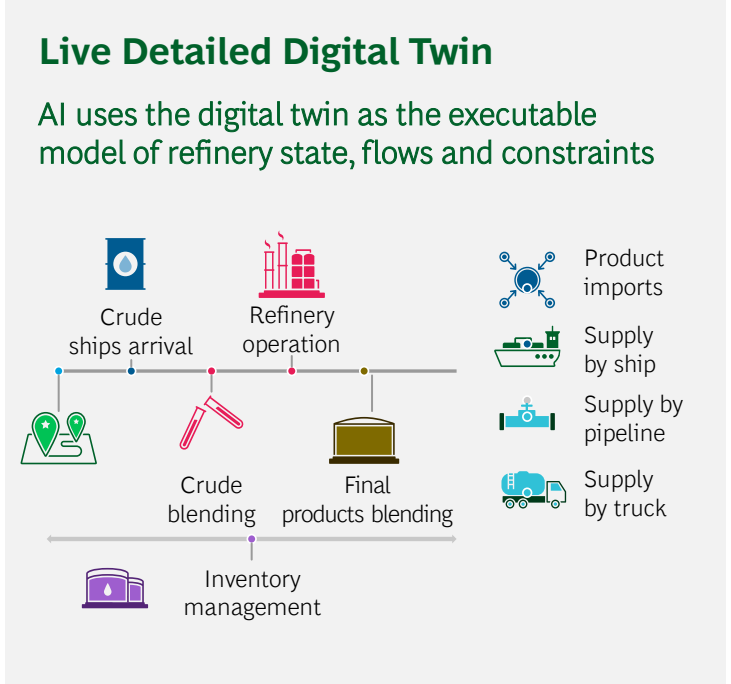
- High complexity across units, tanks, logistics, and time
- Manual/Excel-based scheduling focused on feasibility over value maximization
- Slow response to disruptions leading to margin leakage

The Impact

~0.15 – 0.30 \$/bbl
Margin uplift

 **How AI played a role**

AI scheduler runs a digital twin to dynamically reoptimize short-term refinery operation schedule as operational variability occurs



Source: BCG analysis

AI pinpoints margin leakage and analyzes root causes across the integrated value chain to protect refinery margin

The challenge

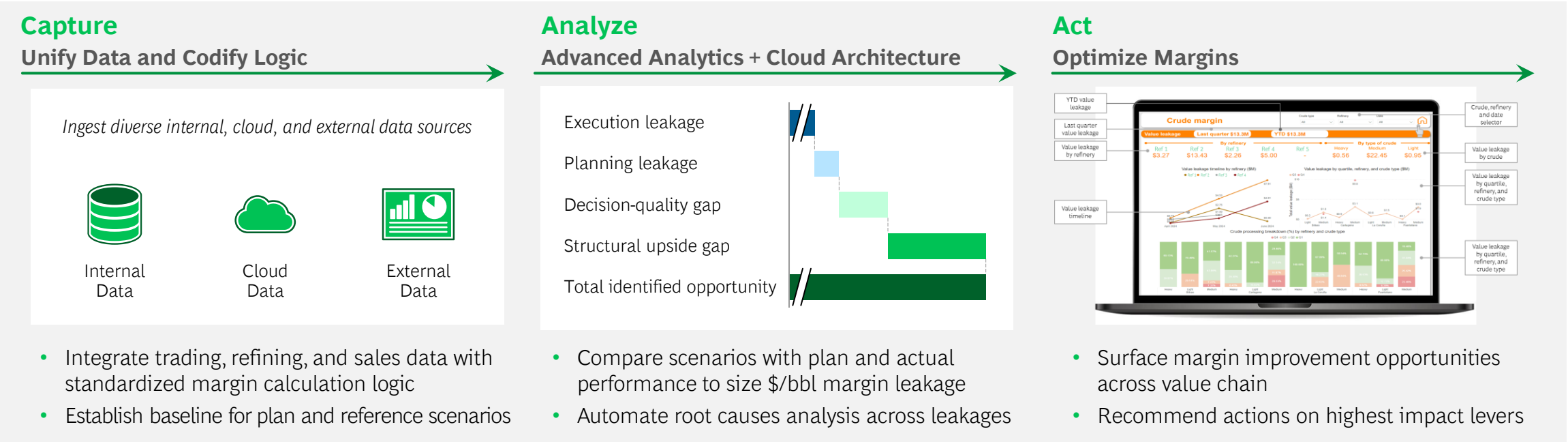
At a European downstream major, operational performance analysis was challenged

- Traditional **backcasting focused on isolated KPIs**, not total margin
- Limited visibility into value leakage** across trading, refining, and sales
- Manual, slow root-cause analysis**

The Impact

~1 – 3 \$/bbl
Identified opportunity¹

 **How AI played a role**
AI detects plan-to-actual margin leakage by comparing actual performance with reference scenarios and surface actionable drivers



Note: 1. Opportunity ranges indicative based on prior implementations. Identified opportunity reflects indicative margin leakage estimated through plan-to-actual and scenario analysis. Realized value will depend on follow-through actions, operational feasibility, and market conditions.
Source: BCG analysis

AI agents orchestrate real-time refinery optimization, translating plant conditions into margin actions for the control room



The challenge

At a Latin American large refiner, operational performance was constrained by:

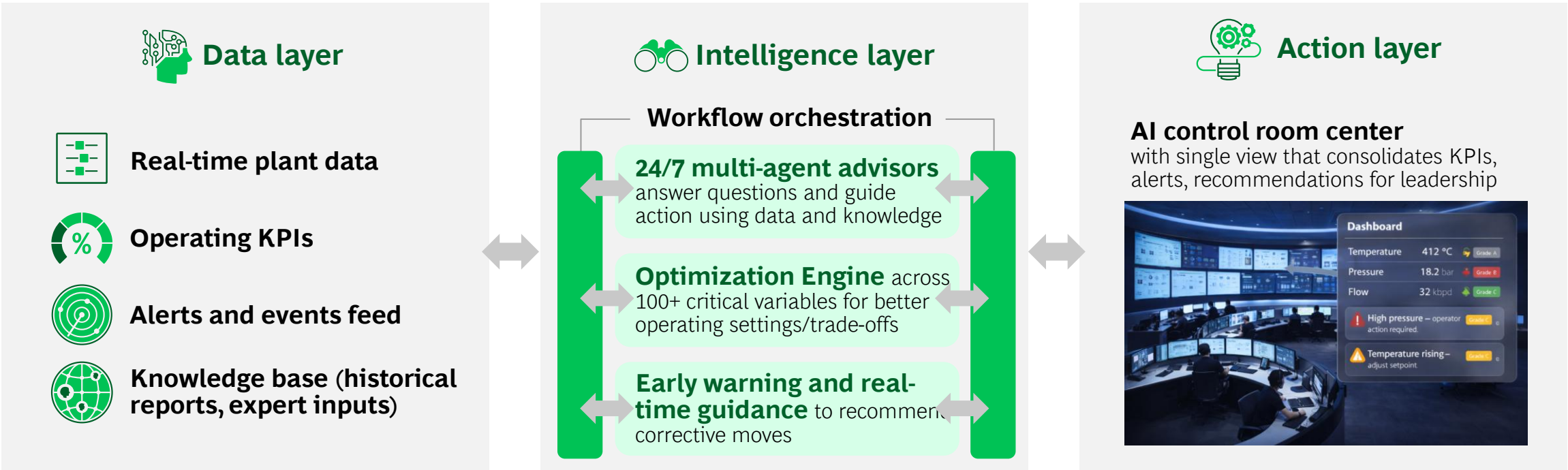
- High operating variability (feed/utility swings, unit constraints)
- Manual, slow diagnosis and response, driving margin leakage
- Decision making was siloed, remaining unit by unit and shift by shift

The impact

+\$80M

 from
~300 kbpd capacity

**How AI played a role**
AI agents orchestrate real-time refinery optimization by combining plant data, KPIs, alerts, and historical knowledge to detect issues early, optimize trade-offs, and guide control room actions.



AI predicts market structure to unlock margin from physical assets by shifting trading from reactive coverage to proactive capture

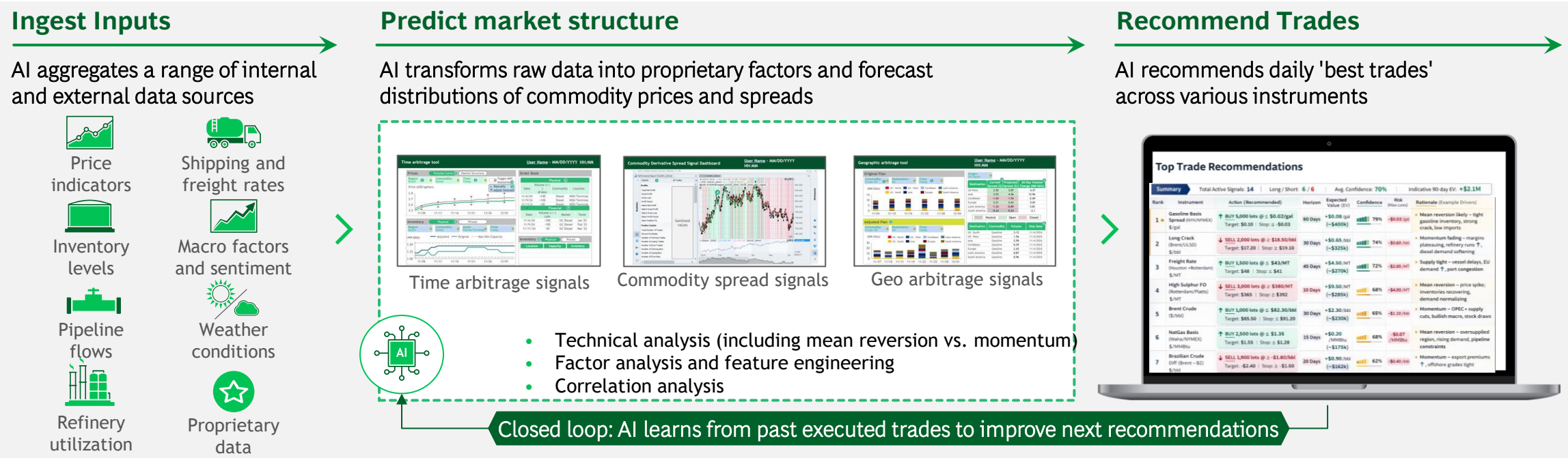
The challenge

- At an American refiner, trading performance was challenged due to:
- Reactive execution focused on near-term cost minimization to meet volume commitments
 - Physical markets difficult to predict outside of 5- to 10-day windows
 - Trader insight/tribal knowledge not codified, making decision logic inconsistent

The Impact

>\$80M+ annual margin opportunity
Ten trades executed in two weeks, with 75% of closed positions cash-positive

 **How AI played a role**
AI forecasts market structure and spread dynamics by combining internal proprietary signals with market feeds



Note: Opportunity ranges indicative based on prior implementations and depend on portfolio, constraints, and market conditions.
Source: BCG analysis

AI agents reduce maintenance costs by orchestrating remote inspections and risk-based interventions across assets

The challenge

At a Southeast Asian NOC’s flagship refinery, asset reliability was challenged due to:

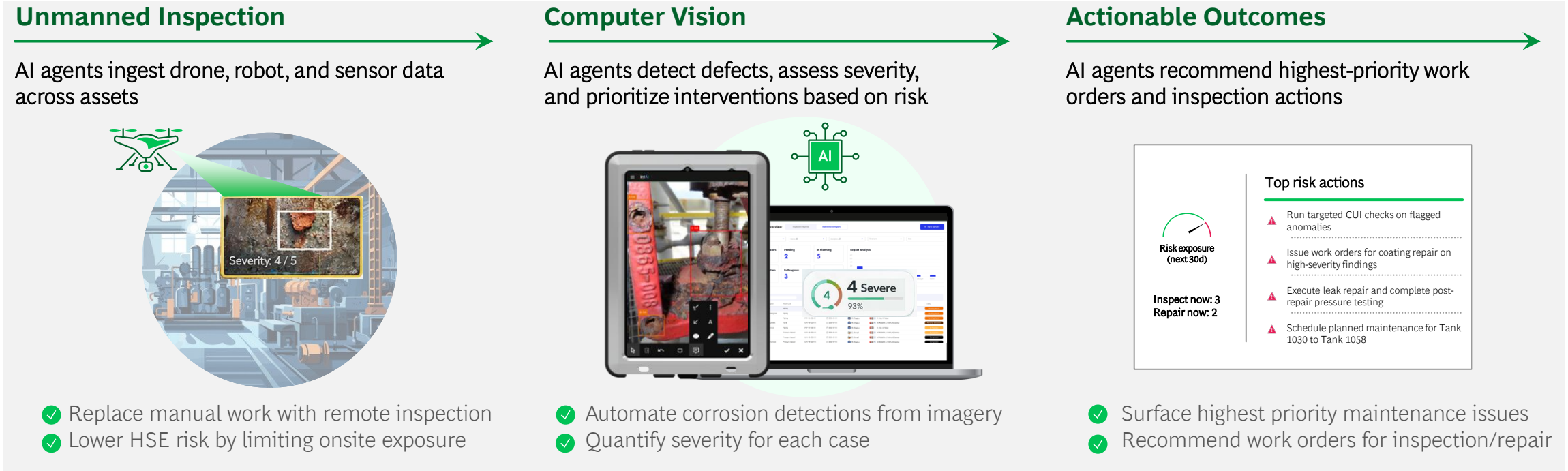
- Maintenance **spend misaligned with actual risk** exposure
- **Limited visibility on failure** probability **vs** economic **impact**
- **Reactive, budget-driven inspection** strategies rather than risk-based prioritization

The Impact

5%-15% Maintenance cost reduction

62% Efficiency gain in inspection workflow

 **How AI played a role**
AI fuses unmanned inspection data and computer-vision integrity analytics to prioritize risk-based inspection and maintenance actions



AI turns static turnaround schedules into continuously reoptimized plans for optimized time, cost, and risks

The challenge

At a Middle Eastern NOC refinery, turnaround and capital schedules were challenged due to

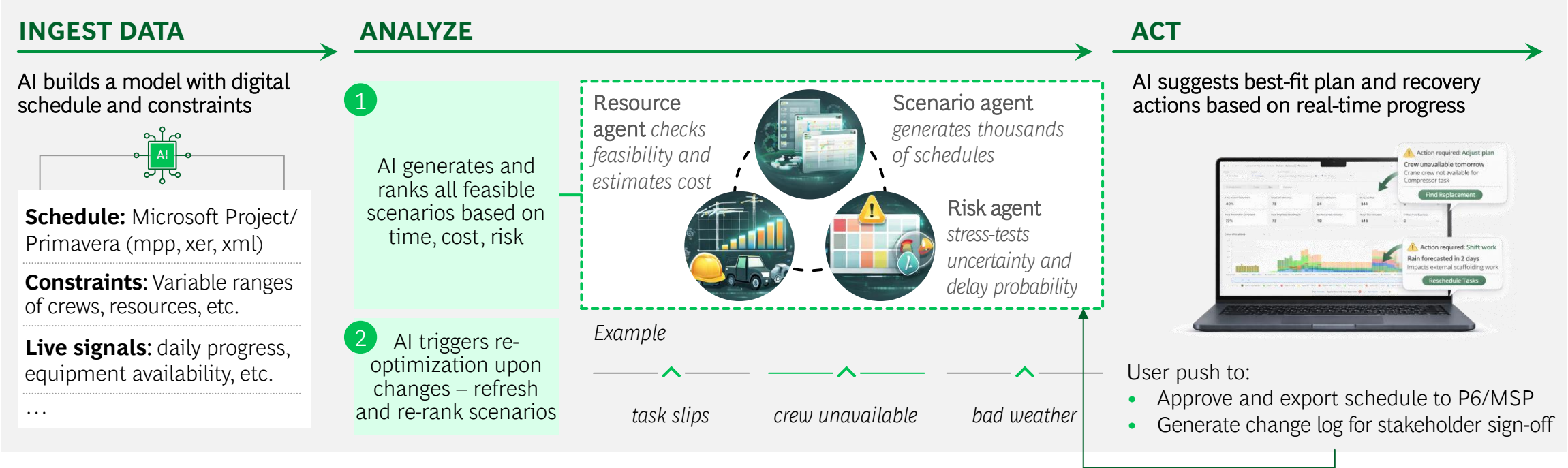
- Manual sequencing struggles with high complexity across work front and dependencies
- Static plans breaking in execution, and slow to reoptimize under changing conditions
- Limited visibility into time-cost-risk trade-offs across scenarios

The impact

~10% cost reduction for routine maintenance

20%-30% reduction in backlog

 **How AI played a role**
AI evaluates thousands of scenarios to recommend the 'best-fit' schedule – and triggers co-pilot reoptimization on disruption



AI improves refinery safety by shifting from reactive incident management to real-time intervention, reducing near-misses and response time

The challenge

At a large Indian refinery, on-site safety and risk management faced gaps due to:

- Reactive incident detection, with issues identified only after escalation
- Manual monitoring unable to keep pace with operational scale and complexity in real time
- Passive CCTV systems, limited to recording without real-time risk detection or analytics

The impact

30%-50% reduction in near-miss incidents

>50% faster response time in intervention

How AI played a role

AI overlays real-time safety detection on CCTV to flag violations, escalate high-risk events, and build an incident log for trend analytics



Run 20+ **real-time safety scenarios** with customizable detection parameters to **detect violations early**



Example of computer vision detection of safety violations at a refinery



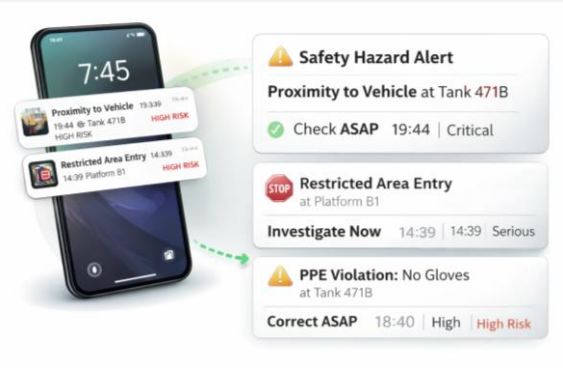
Provide **continuous incident logging and safety trend analytics** to pinpoint hotspots and repeated violations



Incident log and trend analytics dashboard



Trigger **real-time alerts** via mobile notifications and audio alarms to **prompt immediate intervention**



Operator-ready view of active safety scenarios and alerts

Organizational inertia, fragmentation, and weak governance are key implementation hurdles to successful AI transformation in refineries



Fragmented priorities

Diffused priorities and agendas

(across operations, maintenance, sales) with no common view of impact—**dilutes focus and funding**, creating many pilots, but limited scale-up of high-value use cases



Legacy IT systems

Plants operate on **fragmented and aging IT/OT stacks** (e.g., DCS/SCADA, SAP/CMMS) leading to gaps in data quality and integration—**slows deployment of digital and AI use cases** in control room and field



Siloed operations

Refineries make decisions by unit and function **with limited cross-unit and asset coordination**, preventing end-to-end optimization of throughput, margin, and reliability trade-offs



Inflexible operating models

Delivery model is **designed to run day-to-day operations, such as “keeping the lights on.”** Talent and org not optimized with resources dedicated to fast implementation, scaling, and learning



Lack of governance

Dedicated resources and senior sponsorship required to centrally run the transformation, constantly assessing value and steering use case scaling to keep implementations on track and “fund the journey”

Future-built refineries overcome these hurdles through value-driven multiyear transformations

What future-built companies do differently



Pursue a multiyear strategic ambition



Reshape and reinvent for P&L impact



Implement an AI-first operating model



Secure and enable necessary talent



Use fit-for-purpose technology and data

Outperformers vs. laggards

C-suite actively engaged in AI agenda (refinery general manager, functional leaders)

More likely to appoint **chief AI and data officers**

Greater focus **on P&L impact over tool deployment**

Stronger AI **governance and value tracking**

Greater deployment and scaling of AI workflows

Strategic workforce planning for AI in place

Mature and **responsible AI guardrails** and governance

Shared **business-IT ownership** (e.g., engineering and digital) in implementation

Small number of capable AI leaders shaping requirements and vendor choices

Higher business-led ownership of AI solution design and deployment

Greater workforce enablement without building large internal AI teams

Standard templates and enterprise-wide data models

Synergistic, enterprise AI platform as an efficient backbone

AI agents and solutions will enable simplified human-in-loop workflows to provide necessary automation with checks and balances



Example of human-agent¹ supervision flow in refinery operations, such as crude slate, unit optimization

Strategic decision layer (L2)

Human-led

sets crude slate strategy, throughput/yield targets, product quality guardrails, emissions and safety limits

Orchestration layer (L3)

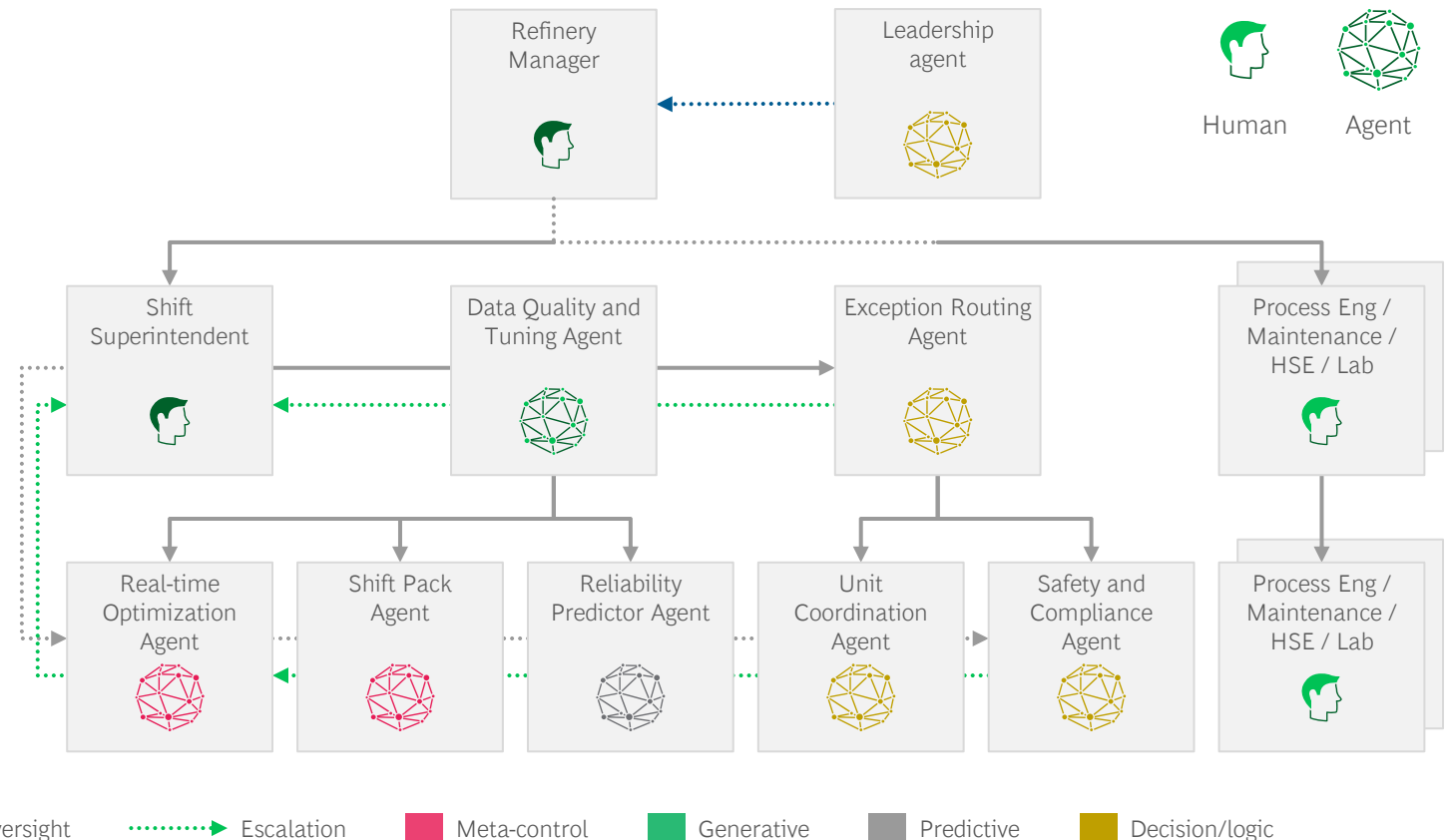
Co-led/split responsibilities

human governs critical decisions such as constraint exceptions and major rate changes; agents handle the baseline by recommending setpoints and actions based on constraints, energy cost, reliability risk

Execution layer (L4)

AI-led

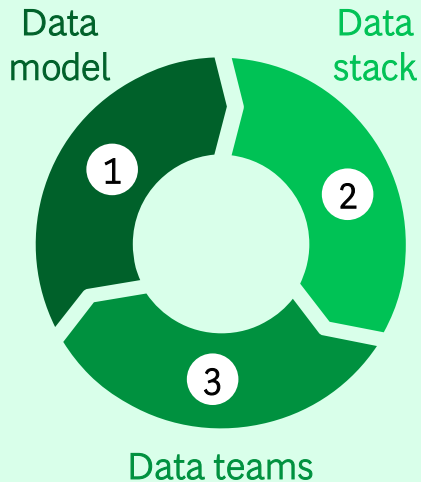
optimize unit operation and energy use, draft shift logs and work orders, monitor alarms and anomalies; detect safety, quality, and compliance triggers



Source: BCG experience

1. Agents connected to systems such as DCS/SCADA | Historian | LIMS | CMMS | LP/Scheduling.

AI-first organizations use business context fabric and easy access to structured and unstructured data to enable development of AI agents



1 Business context fabric

- Agents need more than data, they need **business context**, including:
 - Objectives** – outcomes, not outputs
 - Resources** – tools and data
 - Constraints** – safety and spec limits and regulatory limits

2 Unstructured data ready for agent retrieval

- Enable retrieval over unstructured data** by converting documents (e.g., contracts and network manuals) into searchable formats that agents can retrieve at runtime
- Implement short- and long-term agent memory** (session and episodic or semantic) to retain and reuse interaction history

3 Data products for agents

- APIs and model context protocol** (MCP) that agents and data teams can call autonomously to **expose data**
- Federated data ownership**, with business teams owning data while platforms expose it via APIs for agent consumption
- Data-driven decision-making culture**, with shared playbooks and templates to standardize high-quality agent-supported decisions

Bringing this to life across refineries

- Must have well-connected integration** across lab, plant, quality, and customer systems (e.g., ELN/LIMS, plant sensor, and batch data)

- Must mine thousands of data records daily** to surface relevant insights

- Users must be equipped with intuitive platforms** to interact with and supervise agents

- Continuous monitoring and enhancements** are essential to ensure the capability scales and adapts with the business

Key takeaway: Data must be treated as a critical enterprise asset. Investing in quality, governance, and management is essential for scaling agents across chemical players.

Building an agentic ecosystem requires a new stack on top of traditional business systems and data



Illustrative

Traditional

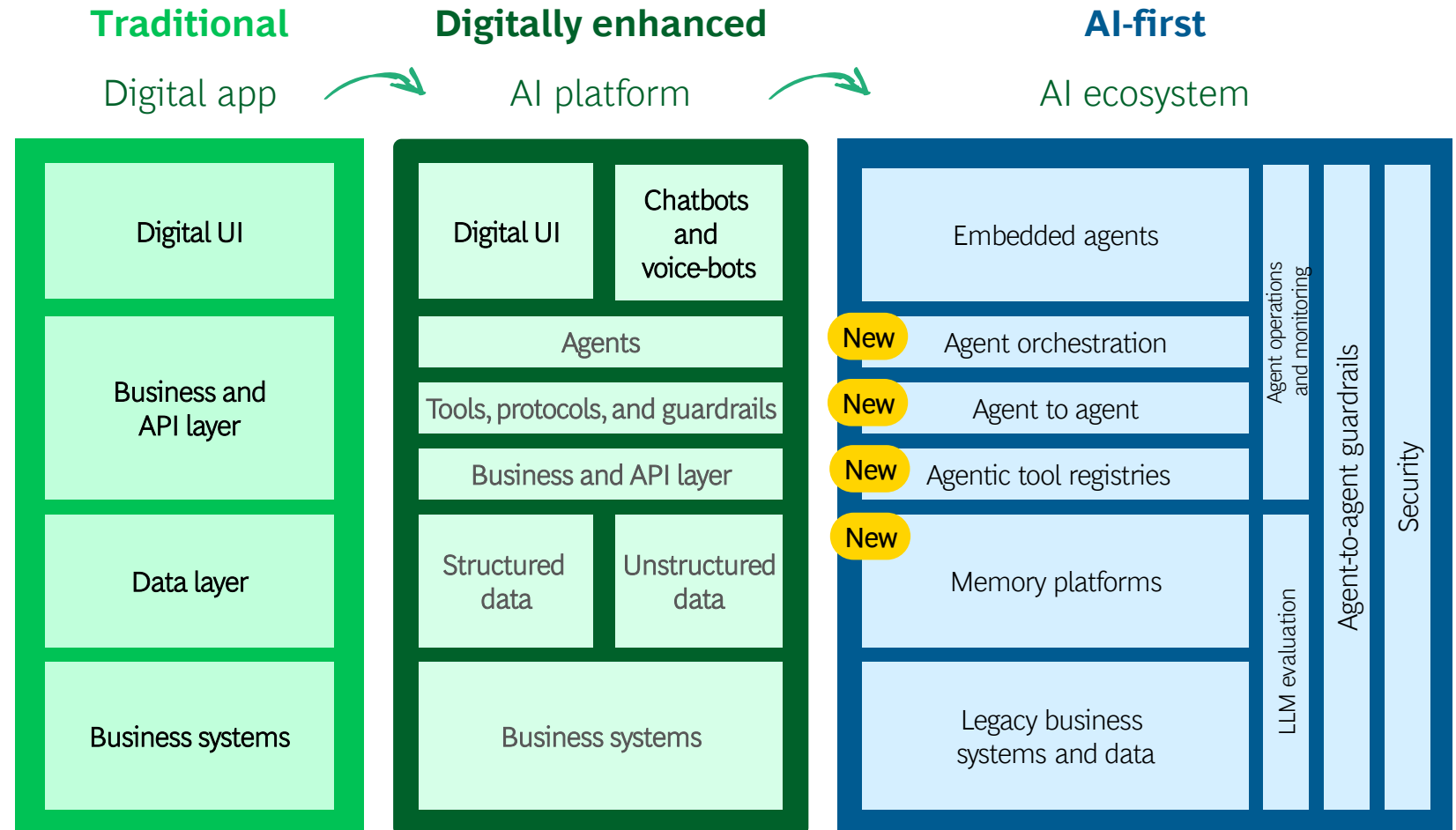
Conventional digital application architecture is organized into distinct layers with limited flexibility and integration resulting in reduced adaptability and greater rigidity

Digitally enhanced

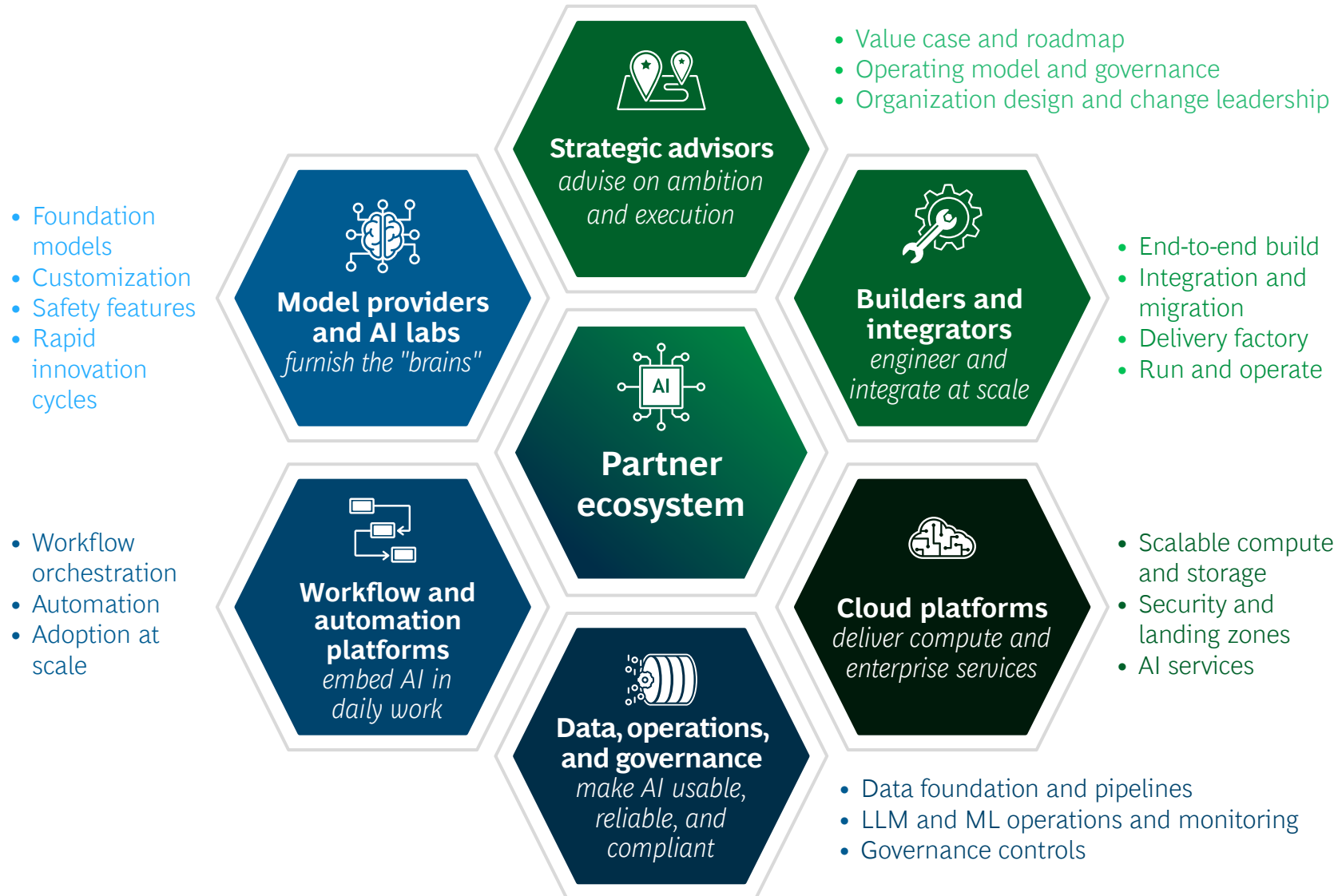
AI-driven architecture combining tools like chatbots, agents, and protocols over traditional structures. Integrates diverse data for enhanced adaptability and efficiency

AI-First

An ecosystem built around embedded AI agents, orchestrations, and tools, emphasizing secure, observable agent-to-agent interactions. Integrates legacy systems and data to boost AI-driven operations.

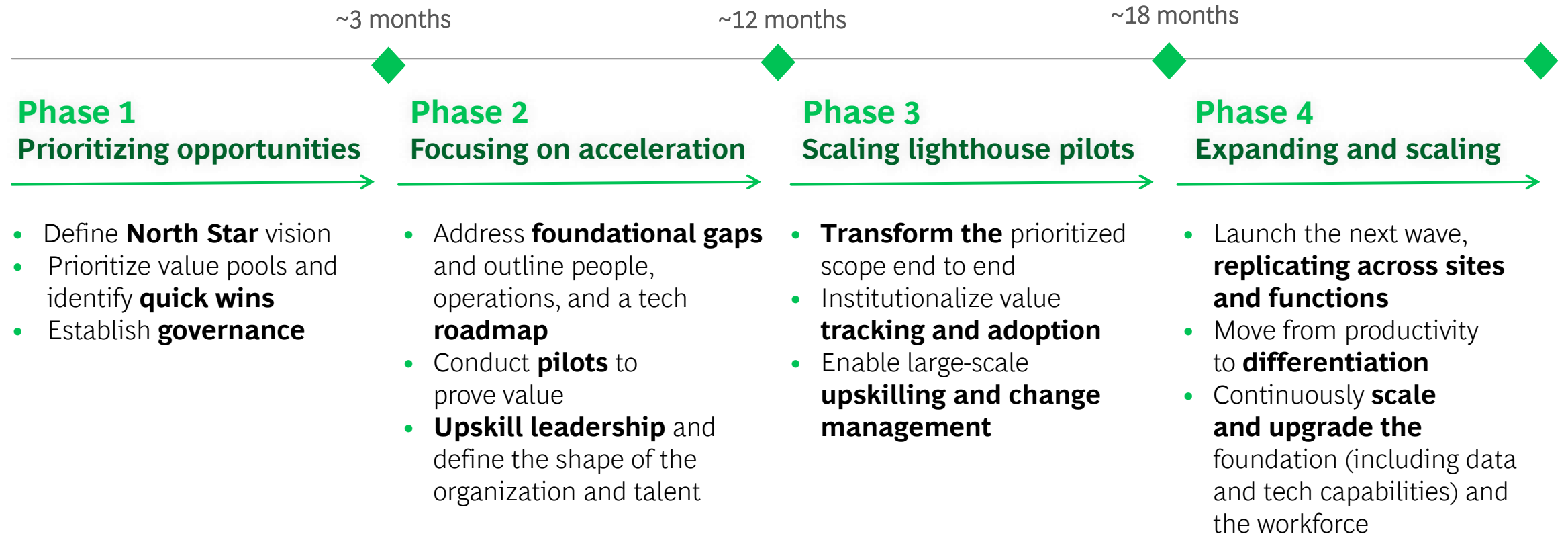


Best-in-class refiners strategically leverage partners to accelerate AI delivery on top of traditional platforms



Value proposition varies across partners. Critical to select best-in-role and orchestrate capabilities across the build-buy-partner mix

Transformation to future-built refineries requires an early focus on prioritized opportunities before expansion and scale-up



Underpin the transformation journey with increasing investments in enterprise foundations including core tech and data, people, and responsible AI

Is my organization on track to be AI future-ready?

Leaders' checklist



Leadership and strategy

set the direction

AI is a strategic priority communicated to the organization

- ☐ I have a named AI sponsor (C-level or board level)
- ☐ AI progress is a **standing agenda item** in leadership meetings



Solutions and business value

determine priorities

We've identified functions or E2E processes where AI can create tangible value

- ☐ We **prioritize** use cases identified for each function, with target outcomes
- ☐ Each use case has a **business sponsor** with P&L accountability
- ☐ We track both business ROI and operational outcomes from AI cases



Funding and investment

back it with resources

Our leadership has a standard way to request AI funding that communicates ROI

- ☐ I have an approved **budget** for AI-ready data, talent, and implementation
- ☐ There is a defined **three-year AI investment envelope** (% of revenue)



Data and technology

build the foundation

Named data owners/stewards exist in business units or functions (e.g., operations, maintenance)

- ☐ We have an **active program** cleaning, governing, and integrating refinery data from control business systems (including DCS/SCADA, APC, MES, CMMS)
- ☐ Our critical data is AI ready (structured and labeled, accessible, compliant)



People and processes

drive adoption

Our leadership team is undergoing AI fluency training

- ☐ We have **cross-functional AI teams** that combine refinery domain experts (field operations, process engineering, maintenance) with OT/IT and Centers of Excellence
- ☐ There is a **change management plan** to address trust, compliance, and regulation
- ☐ We have an **AI talent plan** to attract, retain, and motivate talent



Governance and responsible AI

scale with trust

Wins and lessons learned from pilots communicated visibly across the organization

- ☐ We have an **AI governance council** (ethics, compliance, regulatory).
- ☐ We operate under a **responsible AI framework** (safety, explainability, bias)
- ☐ I receive a **quarterly AI impact report** with business and operational outcomes

BCG experts | Key contacts for refinery AI transformations

Americas



Ilshat
Haris



Oleg
Gubin



John
Florez



Buddy
Myers



Elisabet Assens
Gibert



Wade
Barnes

Europe, Middle East, and Africa



Alex
de Mur



Alberto
de la Fuente



Jaime
Ruiz-Cabrero



Ana María
Martínez



Sébastien
Rexhausen



Arthur
Akhmetov

Asia Pacific



Arun
Rajamani



Asheesh
Sastry



Rahool Pai
Panandiker



Kashmira
Jirafe



Anish
Nazimudeen



Achraf
El Ouali



BCG